

Tarun Harish E

+91 7907272149 | tarunharish2k4@gmail.com | linkedin.com/in/taruntxe | github.com/TarunTXE

SUMMARY

Final-year B.Tech IT student with hands-on experience in full-stack web development (React.js, Node.js, MongoDB, Django) and Machine Learning & Applied AI. Strong foundation in software development, model building, problem-solving, and building practical, real-world applications.

EDUCATION

Institute of Engineering and Technology, University of Calicut
Bachelor of Technology in Information Technology

Malappuram, Kerala
2023 – 2027

EXPERIENCE

BharatCares – IBM SkillsBuild

Aug. 2026 – Present
Remote

Machine Learning and Applied AI Intern

- Completing a 6-week Machine Learning and Applied AI internship in collaboration with AICTE, IBM SkillsBuild, and BharatCares.
- Building and training AI models using IBM watsonx/Watson Studio and Teachable Machine across supervised, unsupervised, deep, and reinforcement learning.
- Developing a project-based ML solution aligned with the UN Sustainable Development Goals (SDGs).

Zoople Technologies

Nov. 2025 – Dec. 2025
Kozhikode, Kerala

Python Django Intern

- Developed database-driven web applications using Python, Django, HTML, CSS, and SQLite.
- Implemented backend functionality and CRUD operations using Django's MVT architecture.
- Debugged and integrated databases across real-world development workflows.

PROJECTS

Signal Hire – Portfolio & Career Platform | *React (Vite), Tailwind CSS, Node.js, Express.js, MongoDB*

- Built a full-stack portfolio builder with customizable templates, a live editor, and responsive previews.
- Implemented JWT-based authentication and secure user sessions.
- Built a peer feedback and rating system, plus a portfolio strength analyzer with improvement suggestions.
- Added one-click PDF resume export and live job search via the JSearch API.

RAG Evaluation Framework – Automated Retrieval & Faithfulness Benchmarking | *Python, FastAPI, ChromaDB, React*

- Built a RAG pipeline with an automated evaluation harness benchmarking retrieval and faithfulness across chunking, embedding, and reranking strategies.
- Built the backend with FastAPI, fastembed embeddings, ChromaDB vector store, and the Groq API for generation.
- Built a React dashboard to visualize benchmark results, with PDF upload for querying custom documents.
- Benchmarked embedding models, identifying a retrieval recall drop from 100% to 83.33% with a weaker model.
- Deployed backend on Render and frontend on Vercel with a live public demo.

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript, SQL

Frontend: HTML5, CSS3, React.js

Backend: Node.js, Express.js, Django, FastAPI, fastembed, Groq API

Databases: MongoDB, SQLite, MySQL, ChromaDB

Tools: Git, GitHub, VS Code, Render, Vercel

ACTIVITIES & CERTIFICATONS

National Service Scheme (NSS) – Volunteer

2023 – 2025

Generative AI in Action – IBM SkillsBuild

Aug. 2026

The Complete Full-Stack Web Development Bootcamp – Udemy

Sept. 2026